



SMART PARKING FINDER

Internship Project Presentation

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Enrollment No.: 25scs1003000912

Program & Semester: B.Tech CSE — 3rd Semester

Organization: DIGISAMAKSH Private Limited

Internship Period: 01 June 2026 – 15 August 2026

Mentor: Mr. Raushan

SOFTWARE ENGINEERING INTERN

Technology & Development Department

Project: Smart Parking Finder

Project concept and functional scope.

- Smart Parking Finder is designed to help users identify suitable parking spaces more efficiently.
- The system focuses on presenting parking information in a simple, organized and user-friendly way.
- Core idea: search for parking → view available/suitable options → compare relevant information → choose a parking option.
- The design can be extended later with live availability, maps and booking.
- This presentation describes the proposed functional design; it does not claim a live deployment unless supported by project evidence.

Problem Statement

Why a smart parking solution is useful.

- Drivers may spend significant time searching for parking in busy areas.
- Information about parking availability may not be easily accessible before reaching a location.
- Repeated searching can increase travel time, fuel consumption and inconvenience.
- A centralized digital interface can make parking information easier to discover and compare.
- The project therefore focuses on improving the parking-discovery experience through software.

Project Objectives

What the Smart Parking Finder aims to achieve.

- Provide a simple interface for discovering parking spaces.
- Organize parking information such as location, availability/status and relevant details.
- Support search and filtering so users can find suitable options more easily.
- Design the application so that real-time data and booking can be added later.
- Apply programming, data-handling, testing and software-development concepts to a practical problem.

Scope of the Project

Current scope and realistic future boundary.

- Parking-space listing and discovery.
- Search/filter functionality based on user requirements.
- Availability/status representation using stored or updateable data.
- User-oriented presentation of parking information.
- Future integration of maps, live occupancy, reservation and digital payment.
- Admin-side management of parking locations and capacity can be added in later versions.

Technologies & Tools

Technologies aligned with the project and internship focus.

- Python — primary programming language used for application logic and data processing.
- HTML & CSS — suitable for structuring and styling the user interface.
- JavaScript — can support client-side interactions where required.
- Database — for storing parking locations, capacity, availability and related records.
- APIs — can be integrated for maps/location services and future real-time functionality.
- Development and testing can be performed using a Python environment, code editor and web browser.

System Architecture

High-level logical architecture.

- User Interface → Application Logic → Data Layer → Parking Results.
- Frontend collects the user's search or parking requirements.
- Python application logic validates the request and processes parking data.
- Data layer stores parking location, capacity and status information.
- The application returns relevant parking options to the user.
- Future modules can connect map APIs, real-time sensors and booking/payment services.

Methodology / Workflow

Development approach followed for the project.

- 1. Requirement analysis — understand the parking problem and user needs.
- 2. Planning — define objectives, scope, features and data requirements.
- 3. Design — prepare application flow and system architecture.
- 4. Development — implement core logic using Python and build the user-facing components.
- 5. Testing — check functional flows, input handling and expected outputs.
- 6. Documentation — record the work, observations and possible improvements.
- 7. Enhancement planning — identify realistic next steps such as maps, live availability and booking.

Key Features

Functional features of the proposed solution.

- Parking-space search and listing.
- Availability/status display.
- Search and filtering concept.
- Clear presentation of parking details.
- Modular design for future expansion.
- Potential booking and map-navigation integration in later versions.

Smart / AI Extension

AI features as future enhancement, not a claim of current implementation.

- Parking-demand prediction can estimate busy periods using historical parking data.
- A recommendation module can rank parking options using factors such as distance, availability and user preference.
- Occupancy prediction can be explored using historical or sensor/camera data, subject to data availability and privacy requirements.
- AI can therefore enhance the basic finder without changing the project's main objective.
- These AI capabilities are proposed future extensions unless separately implemented and demonstrated.

Implementation & Testing

Practical development and quality checks.

- Defined the application's functional requirements and user flow.
- Planned the data required to represent parking locations and availability.
- Implemented/structured the core application logic around Python.
- Tested the logic with representative inputs and expected outputs.
- Reviewed usability, data handling and possible edge cases.
- Documented improvements required for a more complete production-ready system.

Challenges & Solutions

Practical engineering considerations.

- Changing parking availability → use updateable status data and a clear data model.
- Finding relevant parking options → provide search and filtering.
- Keeping the interface simple → prioritize essential parking information.
- Future real-time requirements → keep the architecture modular and API-ready.
- Scaling the project → separate interface, application logic and data components.

Future Scope

Possible next-stage development.

- Real-time parking availability through APIs or IoT sensors.
- GPS/map integration for route guidance.
- Online reservation, cancellation and booking history.
- AI-based parking recommendations and demand forecasting.
- Digital payment integration.
- Admin dashboard for parking operators.
- Analytics for occupancy, peak hours and parking utilization.

Conclusion

Summary.

- The internship provided practical exposure to software development and professional project work.
- Smart Parking Finder addresses a real-world parking-discovery problem through a structured digital solution.
- The project combines Python-based logic, data handling and user-focused application design.
- The architecture provides a foundation for future maps, live availability, booking and intelligent recommendations.
- The internship strengthened technical, documentation, problem-solving and teamwork skills.

Date: 19th August 2026
Employee ID: DIGI/DI/2026/0210

Reference No.: DIGI/IOL/0885



INTERNSHIP COMPLETION CERTIFICATE TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Anushka Baranwal**, s/d/o **Mr.Dinesh Kr. Baranwal**, has successfully completed an internship as **Software Engineering Intern** in the **Technology & Development Department** at **DIGISAMAKSH Private Limited**. The internship tenure was from June 01, 2026 to August 15, 2026.

During the internship period, **Anushka Baranwal** has shown a high level of commitment, technical understanding, and problem-solving ability while working on assigned projects and tasks.

Key Responsibilities & Contributions

- ✚ **Software Development Support:** Assisted in coding, debugging, and testing software applications while ensuring adherence to project guidelines.
- ✚ **Technical Documentation:** Prepared and maintained project-related documentation including code comments, user guides, and technical reports.
- ✚ **Research & Learning:** Conducted research on new technologies, frameworks, and tools to support the ongoing development work of the department.
- ✚ **System Maintenance:** Supported the team in troubleshooting issues, performing system updates, and ensuring smooth functioning of technical infrastructure.
- ✚ **Team Collaboration:** Worked closely with developers, testers, and project managers to complete assigned modules and contribute effectively to project milestones.
- ✚ **Innovation & Problem-Solving:** Suggested improvements in workflows and participated in brainstorming sessions to enhance project outcomes.

Internship Evaluation

Based on the overall performance, learning attitude, and contributions during the internship, **Anushka Baranwal** has been evaluated on the following parameters:

- ✚ **Technical Skills:** Demonstrated good knowledge of IT concepts, coding practices, and problem-solving.
- ✚ **Work Quality:** Delivered assigned tasks with accuracy and within timelines.
- ✚ **Communication Skills:** Maintained professional communication and clarity while coordinating with team members.
- ✚ **Learning & Adaptability:** Showed eagerness to learn new skills and adapt to evolving project requirements.
- ✚ **Professional Conduct:** Maintained discipline, punctuality, and a positive approach throughout the internship.

Overall, **Anushka Baranwal** has successfully met the expectations of the internship program and has added value to the Technology & Development Department of DIGISAMAKSH Private Limited. We wish **Anushka Baranwal** success in all future endeavors.

For

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